

Complex Trigonometric and Related Functions

MA4410

Power Series Definitions (5.4.1)

Definition

For $z \in \mathbb{C}$, define

$$\sin z := \sum_{n=0}^{\infty} (-1)^n \frac{z^{2n+1}}{(2n+1)!}, \quad \cos z := \sum_{n=0}^{\infty} (-1)^n \frac{z^{2n}}{(2n)!}.$$

Both series have infinite radius of convergence and define entire functions.

Complex Exponential Identities

Definition

For $z \in \mathbb{C}$, define

$$\sin z := \frac{e^{iz} - e^{-iz}}{2i}, \quad \cos z := \frac{e^{iz} + e^{-iz}}{2}.$$

These definitions could be used to extend the real sine and cosine using the complex exponential.

Why are these the same

Derivation Sketch

Insert the power series for e^{iz} and e^{-iz} :

$$e^{iz} = \sum_{n=0}^{\infty} \frac{(iz)^n}{n!}, \quad e^{-iz} = \sum_{n=0}^{\infty} \frac{(-iz)^n}{n!}.$$

Separating even and odd powers yields the power series for $\cos z$ and $\sin z$.

Why Power Series First?

- Power series make sense for all $z \in \mathbb{C}$ without prior definitions.
- Infinite radius of convergence avoids domain and branch issues.
- Term-by-term differentiation and algebra are justified.
- Exponential formulas become *consequences*, not definitions.

Basic Properties

For all $z \in \mathbb{C}$:

- $\sin z$ and $\cos z$ are entire functions.
- $\sin(-z) = -\sin z$, $\cos(-z) = \cos z$.
- Euler formulas lead directly to identities.

Example: Evaluating $\sin(ix)$

Let $x \in \mathbb{R}$. Then

$$\sin(ix) = \frac{e^{-x} - e^x}{2i} = i \frac{e^x - e^{-x}}{2} = i \sinh x.$$

Complex sine naturally introduces hyperbolic functions.

Example

Solve $\sin z = 0$.

$$\sin z = 0 \iff e^{iz} = e^{-iz} \iff e^{2iz} = 1.$$

Thus

$$2iz = 2\pi ik \implies z = \pi k, \quad k \in \mathbb{Z}.$$

Same zeros as in the real case.

Definition

For $z \in \mathbb{C}$, define

$$\sinh z := \frac{e^z - e^{-z}}{2}, \quad \cosh z := \frac{e^z + e^{-z}}{2}.$$

These mirror the trigonometric definitions with z replacing iz .

Identities

For all $z \in \mathbb{C}$:

- $\cosh^2 z - \sinh^2 z = 1.$
- $\sinh(iz) = i \sin z.$
- $\cosh(iz) = \cos z.$

Example: $\cos(x + iy)$

Let $x, y \in \mathbb{R}$. Using exponentials,

$$\cos(x + iy) = \cos x \cosh y - i \sin x \sinh y.$$

Real and imaginary parts mix trigonometric and hyperbolic behavior.

- $|\sin(x + iy)|$ and $|\cos(x + iy)|$ grow exponentially as $|y| \rightarrow \infty$.
- Very different from bounded real sine and cosine.
- Important for understanding zeros and contour integrals.

Summary

- Complex trigonometric functions are defined via exponentials.
- Hyperbolic functions arise naturally from imaginary arguments.
- Many familiar identities persist, but behavior of the real axis is different.